

Solar-Powered Home Lighting System

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This project suggests a system that uses solar power and LED lighting technology in a more efficient way. Solar power systems use solar panels to convert the sun's energy directly to electricity. This electricity is stored in the batteries that can supply only DC output. All available

lighting systems, including LED lights, work on AC power. So the DC supply needs to be converted to AC using inverters, which causes some loss of power. The amount of power lost during this process depends on the efficiency of the inverter.

But, if we look closely, in most of the LED lighting systems, the AC is converted back to DC using switched-mode power supplies (SMPS) having an efficiency of around 80 per cent.

This system proposes to use the power from the battery directly for LED lighting, eliminating the need for DC-to-AC conversion by the inverter and AC-to-DC conversion by the LED light's SMPS supply.

Test Points

Test point	Details
TP0	0V, GND
TP1	5V (with respect to TP0)
TP2	BATT1 voltage (with respect to TP0)
TP3	BATT2 voltage (with respect to TP0)
TP4	Solar panel voltage (with respect to TP0)

PARTS LIST

Semiconductors:

- IC1 - PIC16F877A microcontroller
- IC2 - 7805, 5V regulator
- D1-D5 - 1N4001 rectifier diode
- D6-D7 - MBR1635 rectifier diode
- T1-T5 - BC548 npn transistor
- LCD1 - 16x2 LCD
- LOAD - LED panels

Resistors (all 1/4-watt, 45% carbon):

- R1-R5 - 1-kilo-ohm
- R6 - 10-kilo-ohm
- R7 - 100-ohm
- VR1-VR4 - 10-kilo-ohm preset

Capacitors:

- C1, C2 - 22pF ceramic
- C3, C4 - 0.1µF ceramic

Miscellaneous:

- CON1-CON5 - 2-pin terminal connector
- RL1-RL5 - 5V, 1 C/O relay
- S1 - Tactile switch
- S2 - On/off switch
- BATT1, BATT2 - 12V, 80Ah battery
- X1-X11 - 4MHz crystal oscillator
- 2x80W solar panel
- Battery charger

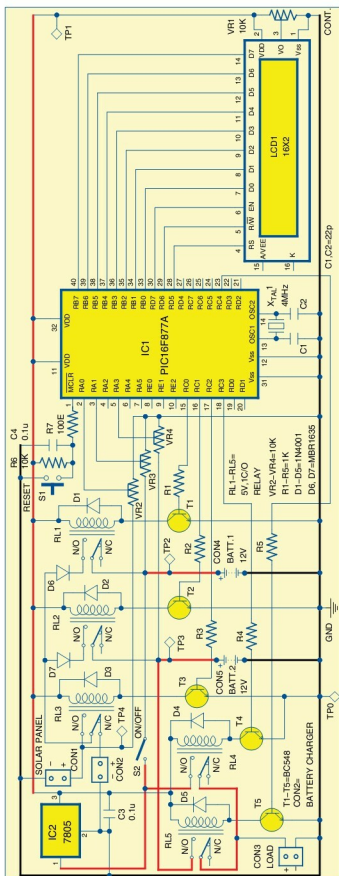


Fig. 1: Circuit diagram of the solar-powered home lighting system